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Week 2 Report

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1. Work Completed This Week

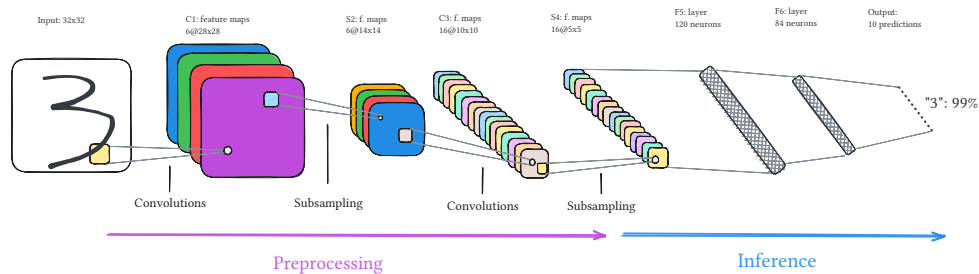
1.1. Groundwork Study (Continued)

Continuing from last week's progress, I spent approximately 15 hours this week on foundational study of the project fundamentals and toolchains.

During this week, I studied:

- **PyTorch Neural Networks:** Running Inference, Backward Propagation using Auto Gradient, Training Batches. [Link to source](#)

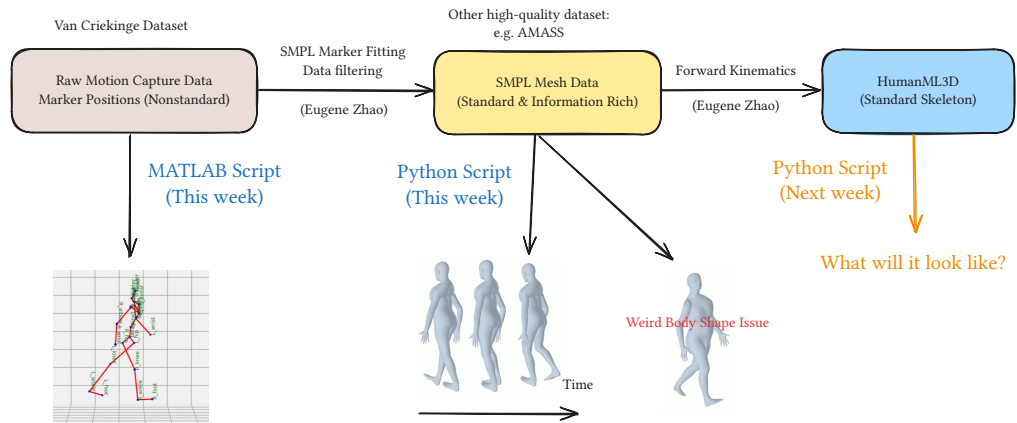
As practice, I trained a simple neural network:



- **Diffusion Models:** Corruption Process, Model Structure, Training Process [Link to source](#)
- **LoRA-MDM Paper:** Briefly analyzed architecture, training dataset requirements, and **results** (*through experimental validation*)

1.2. Analysis of the Van Crieking (VC) Dataset

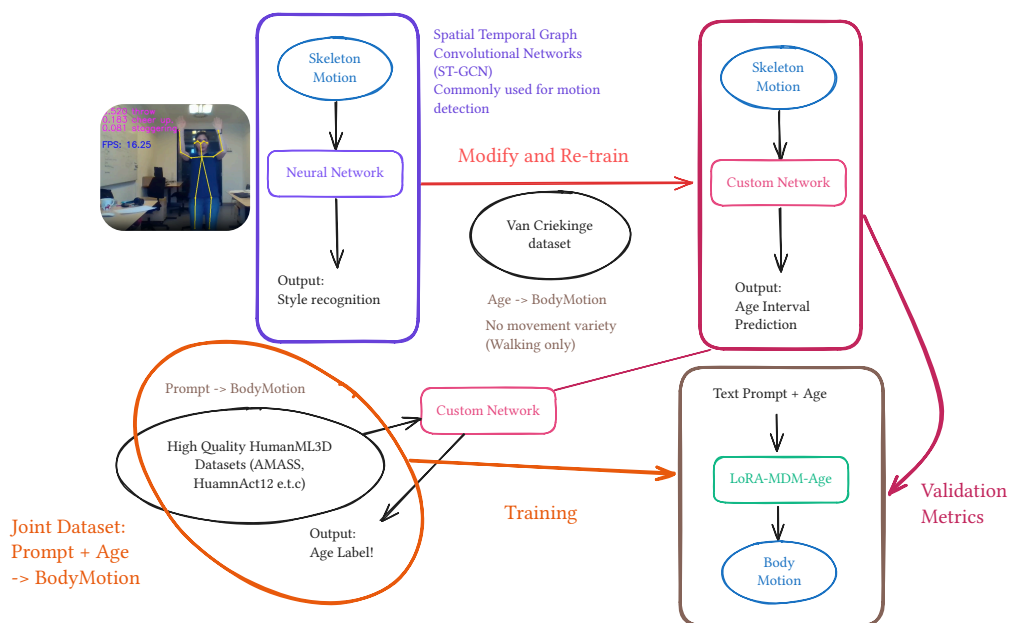
It was suggested that mesh visualization should be used to verify the quality of the processed dataset, so I developed Python/MATLAB scripts for this purpose.



The results show that the data **has multiple issues**. The raw **skeleton** motion is of fine quality, but some files are **corrupted**. Meanwhile, some SMPL mesh data contain an **inflated body shape**. The **first 0.5 seconds of every** motion capture file generate **corrupted mesh data**. Whether this will pollute the final *HumanML3D Standard Skeleton Motion Data* requires further investigation.

1.3. Proposed Method for Age Validation

After the model is trained, the only way to validate its capability to generate age-appropriate motion is to recognize age from the generated motion. The following method is proposed:



2. Project Timeline & Adjustments

Minor adjustments have been made to the project timeline. While unable to explore code implementation details this week, I conducted a detailed analysis of the Van Crieking dataset.

Week 1	Onboarding, foundational study, familiarizing with the project structure (Completed)
Week 2	Complete foundational study; detailed review of existing research (MDM and LoRA-MDM) and their implementations detailed analysis of the Van Crieking dataset and pipeline
Week 3	Start working with the Van Crieking dataset and its processing pipeline; Familiarize with code implementation details; run training experiments
Week 4	Fix all remaining issues with the <i>Van Crieking dataset</i> ; split dataset by age group
Week 5	Train three distinct LoRA adapters: Young, Adult, and Elderly
Feb 15	Based on the rough results generated, submit extended abstract
Week 6–7	Fine-tuning; validation experiments; detailed results and metrics
Week 8–9	Full paper writing ; additional experiments if needed
Week 10 Onwards	Exploratory experiments with new model architecture (Eugene's plan)

Red indicates postponed tasks; **Purple** indicates revised tasks.